Hierarchical Clustering (HCA)

Unsupervised ML technique used to group similar data points into a heirarchy of clusters. It creates a tree-like structure (dendrogram) that represents nested clusters.

↳ Bottom-Up Approach (Agglomerative Hierachical Clustering)

It starts with each data point as an individual cluster and iteratively merges the closest clusters until only one cluster remains or until a specified number of clusters is achieved.

↳ Top - Down Approach (Divisive Hierarchical Clustering)

Starts with one large cluster containing all data points and iteratively splits the clusters into smaller clusters until each data point is its own cluster or until a specified condition is met.

Distance Metrics

$\vec{x}_1 : x_{11}, x_{21} \ldots x_{p1}$

$\vec{x}_2 : x_{12}, x_{22} \ldots x_{p2}$

- Euclidean distance $= \sqrt{\sum_{i=1}^{p} (x_{i1} - x_{i2})^2}$

- Manhattan Distance $= \sum_{i=1}^{p} |x_{i1} - x_{i2}|$

- Mikowski Distance $= \left(\sum_{i=1}^{p} |x_{i1} - x_{i2}|^t \right)^{1/t}$

- Chebyshev's Distance $= \max_i |x_{i1} - x_{i2}|$

- Cosine Similarity $= \dfrac{\sum_{i=1}^{p} x_{i1} * x_{i2}}{\sqrt{\sum_{i=1}^{p} x_{i1}^2} \cdot \sqrt{\sum_{i=1}^{p} x_{i2}^2}}$

- Jaccard Index $= \dfrac{|A \cap B|}{|A \cup B|}$

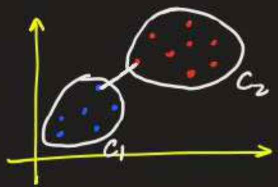
- Hamming distance $= \sum_{i=1}^{n} 1(x_i \neq y_i)$ (# places in which the strings differ)

Linkage Criteria

We know that the distance between two clusters is crucial for hierarchical clustering. There are various ways to calculate the distance between two clusters, and these ways decide the rule for clustering. These measures are called linkage criteria

① <u>Single Linkage</u> : The distance between the closest points of the two clusters

$$D(A, B) = \min\{ d(a_i, b_j) \}$$



eg: A : { (1,2), (1,4) } B : { (2,3), (5,7) }

Calculate the single linkage distance between cluster A and cluster B.

Sol. $a_1 : (1,2)$ $a_2 : (1,4)$ $b_1 : (2,3)$, $b_2 : (5,7)$

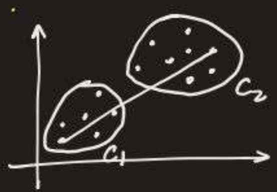
$d(a_1, b_1) = \sqrt{2}$ $d(a_1, b_2) = \sqrt{41}$

$d(a_2, b_1) = \sqrt{2}$ $d(a_2, b_2) = \sqrt{25}$

Single linkage distance = $\boxed{\sqrt{2}}$

② Complete Linkage: The distance between the farthest points of the two clusters.

$$D(A,B) = \max\{d(a_i, b_j)\}$$



eg. A: $\{(1,2), (1,4)\}$ B: $\{(2,3), (5,7)\}$

Calculate the complete linkage distance between cluster A and cluster B.

Sol. $a_1 : (1,2)$ $a_2 : (1,4)$ $b_1 : (2,3)$, $b_2 : (5,7)$

$d(a_1, b_1) = \sqrt{2}$ $d(a_1, b_2) = \sqrt{41}$

$d(a_2, b_1) = \sqrt{2}$ $d(a_2, b_2) = \sqrt{25}$

Complete linkage distance = $\boxed{\sqrt{41}}$

③ **Average Linkage** : The average distance between every pair of points in the two clusters.

$$D(A,B) = \frac{1}{|A| \cdot |B|} \sum_{a \in A} \sum_{b \in B} d(a,b)$$

eg. A : $\{(1,2), (1,4)\}$ B : $\{(2,3), (5,7)\}$

Calculate the average linkage distance between cluster A and cluster B.

Sol. $a_1 : (1,2)$ $a_2 : (1,4)$ $b_1 : (2,3)$, $b_2 : (5,7)$

$d(a_1, b_1) = \sqrt{2}$ $d(a_1, b_2) = \sqrt{41}$

$d(a_2, b_1) = \sqrt{2}$ $d(a_2, b_2) = \sqrt{25}$

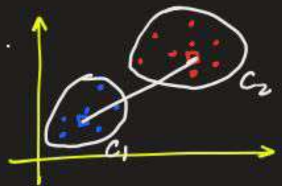
Average Linkage distance $= \dfrac{\sqrt{2} + \sqrt{2} + \sqrt{41} + \sqrt{25}}{4} \simeq \boxed{3.56}$

④ **Centroid Linkage :** The distance between the centroids of the two clusters

$$d(A, B) = d(\bar{a}, \bar{b})$$

$\bar{a}$: centroid of cluster A

$\bar{b}$: Centroid of cluster B



eg. A : $\{(1,2), (1,4)\}$ B : $\{(2,3), (5,7)\}$

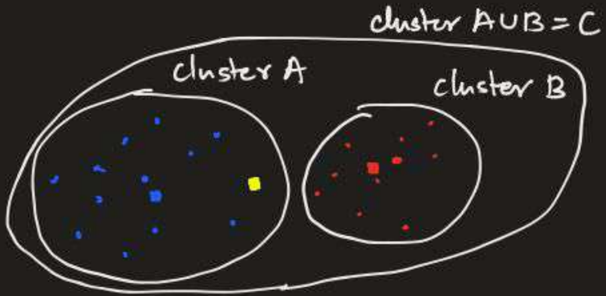
Calculate the centroid linkage distance between cluster A and cluster B.

Sol. Centroid of cluster A : $\bar{A} = \left(\frac{1+1}{2}, \frac{2+4}{2}\right) = (1, 3)$

Centroid of cluster B : $\left(\frac{2+5}{2}, \frac{3+7}{2}\right) = (3.5, 5)$

Centroid linkage distance $= \sqrt{(2.5)^2 + (2)^2} \approx \boxed{3.2}$

(5) **Ward's Linkage**: Minimizes the total within-cluster variance when merging two clusters. It tends to create clusters of small size.



$$W(A) = \sum_{x \in A} \|x - \mu_A\|^2$$

$$W(C) = \sum_{x \in A \cup B} \|x - \mu_C\|^2$$

$$W(B) = \sum_{x \in B} \|x - \mu_B\|^2$$

Wards Criteria: Wards distance between two clusters is the increase in the variance when merging clusters A and B

$$\Delta W = W(C) - (W(A) + W(B))$$

$$= \frac{|A| \cdot |B|}{|A| + |B|} \cdot \|\mu_A - \mu_B\|^2$$

$\|\mu_A - \mu_B\|$: Distance b/w centroids of A and B

Proof: $\mu_C = \dfrac{|A| \cdot \mu_A + |B| \cdot \mu_B}{|A| + |B|}$

$$W(C) = \sum_{x \in A \cup B} \|x - \mu_C\|^2 = \sum_{x \in A} \|x - \mu_C\|^2 + \sum_{x \in B} \|x - \mu_C\|^2$$

$$\sum_{x \in A} \|x - \mu_C\|^2 = \sum_{x \in A} \left\| x - \frac{|A| \cdot \mu_A + |B| \cdot \mu_B}{|A| + |B|} \right\|^2$$

$$= \sum_{x \in A} \left\| x - \frac{(|A| + |B|)\mu_A - |B|\mu_A + |B|\mu_B}{|A| + |B|} \right\|^2$$

$$= \sum_{x \in A} \| (x - \mu_A) + \frac{|B|}{|A|+|B|} \cdot (\mu_B - \mu_A) \|^2$$

$$= \sum_{x \in A} \|(x-\mu_A)\|^2 + 2 \cdot \frac{|B|}{|A|+|B|} \cdot (\mu_B - \mu_A) \boxed{\sum_{x \in A} (x - \mu_A)} + \left\| \frac{|B|}{|A|+|B|} \cdot (\mu_B - \mu_A) \right\|^2 \sum_{x \in A} 1$$

$$= W(A) + \frac{|A| \cdot |B|^2}{(|A|+|B|)^2} \| \mu_B - \mu_A \|^2$$

Similarly

$$\sum_{x \in B} \|x - \mu_C\|^2 = W(B) + \frac{|B| \cdot |A|^2}{(|A|+|B|)^2} \|\mu_A - \mu_B\|^2$$

$$W(C) = W(A) + W(B) + \frac{|A| \cdot |B|^2 + |B| \cdot |A|^2}{(|A|+|B|)^2} \|\mu_A - \mu_B\|^2$$

$$W(C) = W(A) + W(B) + \frac{|A| \cdot |B|}{(|A|+|B|)^2} \|\mu_A - \mu_B\|^2$$

$$\therefore \quad \Delta W = W(C) - (W(A) + W(B)) = \frac{|A| \cdot |B|}{(|A| + |B|)^2} \| \mu_A - \mu_B \|^2$$

eg. A : $\{ (1,2), (1,4) \}$ B : $\{ (2,3), (5,7) \}$

Calculate the ward's linkage distance between cluster A and cluster B.

Sol. $\mu_A = (1,3)$, $\mu_B = (3.5, 5)$

Ward's linkage distance = $\Delta W = \frac{|A| \cdot |B|}{|A| + |B|} \| \mu_A - \mu_B \|^2$

$$= \frac{2 \times 2}{2+2} \left(\sqrt{(2.5)^2 + (2)^2} \right)^2$$

$$= \boxed{10.25}$$